COSTELLO COLLEGE OF BUSINESS
George Mason University

Empirical Asset Pricing

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FNAN 883, Fall 2026

Classes and Contact

Classes	Tuesday, 10:30-13:10
Class meetings location	IN 139
Office	Enterprise Hall 329
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Office Hours	By appointment

Course Description

This course is about efficient markets and empirical asset pricing. We start with reviewing asset pricing theories, asset pricing models, and return predictability. We then focus on cross-sectional asset pricing tests and review the standard methodologies proposed in the literature. We study tests of the CAPM and multi-factor models, including time-series regressions, cross-sectional regressions and GMM. Part of class time is also dedicated to the construction of factors and portfolios for testing. In the second part of the course, we review the applications of asset pricing models to mutual funds, hedge funds and event studies. We will complete the course with a brief review of behavioral finance.

The class will focus on in-class presentations by students, which will constitute 60% of the grade

Course Objectives

1. Develop skills to work with data and conduct empirical analysis in asset pricing
2. Gain the skills and tools to make professional presentations for an academic audience

3. Be able to critically evaluate contributions to the discipline in working and published academic papers.
4.

Textbooks

John Y. Campbell, Andrew W. Lo, and A. Craig MacKinlay. *The Econometrics of Financial Markets*. Princeton, NJ: Princeton University Press, 1997.

John H. Cochrane. *Asset Pricing*. Revised. Princeton, New Jersey: Princeton University Press, 2005. ISBN: 978-0691121376.

Richard H. Thaler, ed. *Advances in Behavioral Finance, Volume II*. Princeton: Princeton University Press, 2005. ISBN: 9781400829125. DOI: doi:10.1515/9781400829125. URL: https://doi.org/10.1515/9781400829125.

Course Contents and Tentative Schedule

The table below presents the topics covered in each week of the class

Date	Topic	Readings/Assignments
Wk 1	Static Asset Pricing Models	
Wk 2		
Wk 3	Market Efficiency and Asset Pricing Tests	
Wk 4		
Wk 5	Empirical Methods	Brennan, Chordia, and Subrahmanyam (1998), Daniel, Grinblatt, et al. (1997)
Wk 6		
Wk 7	Anomalies	
Wk 8		

Wk 9	Mutual Funds and Hedge Funds
Wk 10	
Wk 11	Event Studies
Wk 12	
Wk 13	Behavioral Finance
Wk 14	

Presentation Rubric

	Needs Improvement	Fair	Good	Excellent
Calculations (25%)	Only some calculations are accurate and/or excess ambiguity exists in either steps taken to arrive at mathematical answers and/or methods selected to find solutions.	Most calculations performed with adequate accuracy but ambiguity exists in calculations. More precise and/or detailed execution needed.	Most/majority of calculations performed with accuracy and completeness. Steps are conveyed in a sufficient manner. Little or no ambiguity in mathematical calculations	All calculations performed with accuracy and completeness. All steps are conveyed to arrive at solutions. No ambiguity in mathematical calculations and/or steps taken
Technical Presentation (25%)	Graphs and/or tables may be inaccurate and/or not informative. Graphing/display needs some of the following: title, proper styles and coloring, axis magnitudes, tick values/marks, x-label, y-label, annotation text, and points on the graph/table.	Graphs and/or tables are somewhat accurate and informative. Graphing/display includes some of the following: title, proper styles and coloring, axis magnitudes, tick values/marks, x-label, y-label, annotation text, and points on the graph/table.	Graphs and/or tables are accurate and informative. Graphing/display includes most of the following: title, proper styles and coloring, axis magnitudes, tick values/marks, x-label, y-label, annotation text, and points on the graph/table.	Graphs and/or tables are highly accurate and informative. Graphing/display includes title, proper styles and coloring, axis magnitudes, tick values/marks, x-label, y-label, annotation text, and points on the graph/table.
Technical Explanations (25%)	Interpretation of results in written format is only somewhat conveyed. Calculations, graphs, takeaways, and/or strategic or mathematical relationships may be ambiguous in explanations provided.	Interpretation of results in written format is mostly conveyed. Calculations, graphs, takeaways, and/or strategic or mathematical relationships can be mostly understood from explanations provided.	Interpretation of results in written format conveyed adequately. Calculations, graphs, takeaways, and/or strategic or mathematical relationships can be understood from explanations provided.	Interpretation of results in written format is very clearly conveyed. Calculations, graphs, takeaways, and/or strategic or mathematical relationships can be distinctly understood from explanations provided.
Overall Presentation and Strategy (25%)	Recommendations provided and strategic importance of the assignment is not clearly conveyed in the assignment submission. Evidence provided and connections are either not attempted or not clearly made between calculations and/or theory to provide practical application for investors to utilize.	Recommendations provided and strategic importance of the assignment is mostly available in assignment submission. Evidence and connections are made and/or attempted between calculations and/or theory to practical application for investors to utilize.	Recommendations provided and strategic importance of the assignment is available in the assignment submission. Evidence is and connections can be made between calculations and/or theory to practical application for investors to utilize.	Recommendations provided and strategic importance of the assignment is well crafted and on display in the assignment submission. Evidence is provided and connections are clearly made from calculations and/or theory to practical application for investors to utilize.

Static Asset Pricing Models

1 William F. Sharpe. "Capital Asset Prices: A Theory of Market Equilibrium". In: *Journal of Finance* 19 (1964), pp. 425–442.

2 Eugene F. Fama and Kenneth R. French. "The Cross-Section of Expected Stock Returns". In: *Journal of Finance* 47.2 (June 1992), pp. 427–465.

3 Eugene F. Fama and Kenneth R. French. "Common Risk Factors in the Returns on Stocks and Bonds". In: *Journal of Financial Economics* 33.1 (Feb. 1993), pp. 3–56.

4 Eugene F. Fama and Kenneth R. French. "Multifactor Explanations of Asset Pricing Anomalies". In: *Journal of Finance* 51.1 (Mar. 1996), pp. 55–84.

5 Eugene F. Fama and Kenneth R. French. "A Five-Factor Asset Pricing Model". In: *Journal of Financial Economics* 116.1 (Sept. 2015), pp. 1–22.

6 Robert F. Stambaugh and Yu Yuan. "Mispricing Factors". In: *Review of Financial Studies* 30.4 (Apr. 2017), pp. 1270–1315.

7 Eugene F. Fama and Kenneth R. French. "Choosing factors". In: *Journal of Financial Economics* 128.2 (May 2018), pp. 234–252.

8 Kent Daniel, David Hirshleifer, and Lin Sun. "Short- and Long-Horizon Behavioral Factors". In: *The Review of Financial Studies* 33.4 (Apr. 2020), pp. 1673–1736.

9 Michael Cooper, Huseyin Gulen, and Mihai Ion. "The use of asset growth in empirical asset pricing models". In: *Journal of Financial Economics* 151 (Jan. 2024), p. 103746.

Market Efficiency and Asset Pricing Tests

10 Matthew Richardson and James H. Stock. "Drawing inferences from statistics based on multiyear asset returns". In: *Journal of Financial Economics* 25.2 (Dec. 1989), pp. 323–348.

11 Eugene F. Fama. "Efficient Capital Markets: II". In: *Journal of Finance* 46.5 (Dec. 1991), pp. 1575–1617.

12 Narasimhan Jegadeesh and Sheridan Titman. "Profitability of Momentum Strategies: An Evaluation of Alternative Explanations". In: *Journal of Finance* 56.2 (2001), pp. 699–720.

13 Tarun Chordia and Lakshmanan Shivakumar. "Earnings and Price Momentum". In: *Journal of Financial Economics* 80.3 (2006), pp. 627–656.

14 Kent Daniel and Sheridan Titman. "Market Reactions to Tangible and Intangible Information". In: *The Journal of Finance* 61.4 (Aug. 2006), pp. 1605–1643.

15 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Momentum and Credit Rating". In: *Journal of Finance* 62.5 (Oct. 2007), pp. 2503–2520.

16 Jacob Boudoukh, Matthew Richardson, and Robert F. Whitelaw. "The Myth of Long-Horizon Predictability". In: *The Review of Financial Studies* 21.4 (July 2008), pp. 1577–1605.

17 Michael J. Cooper, Huseyin Gulen, and Michael J. Schill. "Asset Growth and the Cross-Section of Stock Returns". In: *Journal of Finance* 63.4 (Aug. 2008), pp. 1609–1651.

18 Jeffrey Pontiff and Artemiza Woodgate. "Share Issuance and Cross-sectional Returns". In: *Journal of Finance* 63.2 (Apr. 2008), pp. 921–945.

19 Clifford S. Asness, Tobias J. Moskowitz, and Lasse H. Pedersen. "Value and Momentum Everywhere". In: *Journal of Finance* 68.3 (June 2013), pp. 929–985.

20 Kewei Hou, Haitao Mo, Chen Xue, and Lu Zhang. "Which Factors?" In: *Review of Finance* 23.1 (Feb. 2019), pp. 1–35. DOI: 10.1093/rof/rfy032. URL: https://doi.org/10.1093/rof/rfy032.

Empirical Methods

21 Eugene F. Fama and James D. MacBeth. "Risk, Return, and Equilibrium: Empirical Tests". In: *Journal of Political Economy* 81 (1973), pp. 607–636.

22 Douglas T. Breeden and Robert H. Litzenberger. "Prices of State-Contingent Claims Implicit in Option Prices". In: *The Journal of Business* 51.4 (Oct. 1978), pp. 621–651.

23 Douglas T. Breeden, Michael R. Gibbons, and Robert H. Litzenberger. "Empirical Test of the Consumption-Oriented CAPM". In: *Journal of Finance* 44.2 (June 1989), pp. 231–262.

24 Michael R. Gibbons, Stephen A. Ross, and Jay Shanken. "A Test of the Efficiency of a Given Portfolio". In: *Econometrica* 57.5 (Sept. 1989), pp. 1121–1152.

25 Michael J. Brennan, Tarun Chordia, and Avanidhar Subrahmanyam. "Alternative factor specifications, security characteristics, and the cross-section of expected stock returns". In: *Journal of Financial Economics* 49.3 (Sept. 1998), pp. 345–373.

26 Martin Lettau and Sydney L. Ludvigson. "Resurrecting the (C)CAPM: A Cross-Sectional Test When Risk Premia Are Time-Varying". In: *Journal of Political Economy* 109.6 (2001), pp. 1238–1287.

27 Doron Avramov and Tarun Chordia. "Asset Pricing Models and Financial Market Anomalies". In: *Review of Financial Studies* 19.3 (2006), pp. 1001–1040.

28 Jonathan W. Lewellen and Stefan Nagel. "The Conditional CAPM Does Not explain Asset-Pricing Anomalies". In: *Journal of Financial Economics* 82.2 (Nov. 2006), pp. 289–314.

29 Guanhao Feng, Stefano Giglio, and Dacheng Xiu. "Taming the Factor Zoo: a Test of New Factors". In: *Journal of Finance* 75.3 (June 2020), pp. 1327–1370.

30 Theis Ingerslev Jensen, Bryan Kelly, Semyon Malamud, and Lasse Heje Pedersen. "Machine Learning and the Implementable Efficient Frontier". In: *The Review of Financial Studies* (Mar. 2026), hhag022.

Anomalies

31 Narasimhan Jegadeesh and Sheridan Titman. "Returns to buying winners and selling losers: implications for stock market efficiency". In: *Journal of Finance* 48.1 (Mar. 1993), pp. 65–91.

32 Narasimhan Jegadeesh and Sheridan Titman. "Overreaction, Delayed Reaction, and Contrarian Profits". In: *Review of Financial Studies* 8.4 (Jan. 1995), pp. 973–993.

33 Ilia D. Dichev. "Is the Risk of Bankruptcy a Systematic Risk?" In: *Journal of Finance* 53.3 (1998), pp. 1131–1147.

34 Karl B. Diether, Christopher J. Malloy, and Anna Scherbina. "Difference of Opinion and the Cross Section of Stock Returns". In: *Journal of Finance* 57.5 (2002), pp. 2113–2141.

35 Andrew Ang, Robert J. Hodrick, Yuhang Xing, and Xiaoyan Zhang. "The Cross-Section of Volatility and Expected Returns". In: *Journal of Finance* 61.1 (Feb. 2006), pp. 259–299.

15 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Momentum and Credit Rating". In: *Journal of Finance* 62.5 (Oct. 2007), pp. 2503–2520.

36 John Y. Campbell, Jens Hilscher, and Jan Szilagyi. "In Search of Distress Risk". In: *Journal of Finance* 63.6 (Dec. 2008), pp. 2899–2939.

37 Eugene F. Fama and Kenneth R. French. "Dissecting Anomalies". In: *Journal of Finance* 63.4 (Aug. 2008), pp. 1653–1678.

38 Doron Avramov, Tarun Chordia, Gergana Jostova, and Alexander Philipov. "Dispersion in Analysts' Earnings Forecasts and Credit Rating". In: *Journal of Financial Economics* 91.1 (Jan. 2009), pp. 83–101.

39 Gergana Jostova, Stanislava Nikolova, Alexander Philipov, and Christof Stahel. "Momentum in corporate bond returns". In: *Review of Financial Studies* 26.7 (May 2013), pp. 1649–1693.

40 Kewei Hou, Chen Xue, and Lu Zhang. "Digesting Anomalies: An Investment Approach". In: *Review of Financial Studies* 28.3 (2015), pp. 650–705.

Mutual Funds and Hedge Funds

41 Mark Grinblatt, Sheridan Titman, and Russ Wermers. "Momentum Investment Strategies, Portfolio Performance, and Herding: A Study of Mutual Fund Behavior". In: *The American Economic Review* 85.5 (Dec. 1995), pp. 1088–1105.

42 Mark M. Carhart. "On Persistence in Mutual Fund Performance". In: *Journal of Finance* 52.1 (Mar. 1997), pp. 57–82.

43 Kent Daniel, Mark Grinblatt, Sheridan Titman, and Russ Wermers. "Measuring Mutual Fund Performance with Characteristic-Based Benchmarks". In: *Journal of Finance* 52.3 (July 1997), pp. 1035–1058.

44 William Fung and David A. Hsieh. "Empirical Characteristics of Dynamic Trading Strategies: The Case of Hedge Funds". In: *Review of Financial Studies* 10.2 (Summer 1997), pp. 275–302.

45 Bing Liang. "Hedge Funds: The Living and theDead". In: *Journal of Financial and Quantitative Analysis* 35.3 (Sept. 2000), pp. 309–326.

46 Vikas Agarwal and Narayan Y. Naik. "Risks and Portfolio Decisions Involving Hedge Funds". In: *The Review of Financial Studies* 17.1 (Jan. 2004), pp. 63–98.

47 Jonathan B. Berk and Richard C. Green. "Mutual Fund Flows and Performance in Rational Markets". In: *Journal of Political Economy* 112.6 (Dec. 2004), pp. 1269–1295.

48 Mila Getmansky, Andrew W. Lo, and Igor Makarov. "An Econometric Model of Serial Correlation and Illiquidity in Hedge Fund Returns". In: *Journal of Financial Economics* 74.3 (Dec. 2004), pp. 529–609.

49 Anurag Gupta and Bing Liang. "Do hedge funds have enough capital? A value-at-risk approach". In: *Journal of Financial Economics* 77.1 (July 2005), pp. 219–253.

50 Eugene F. Fama and Kenneth R. French. "Luck versus Skill in the Cross-Section of Mutual Fund Returns". In: *The Journal of Finance* 65.5 (Sept. 2010), pp. 1915–1947.

51 Mark Grinblatt, Gergana Jostova, Lubomir Petrasek, and Alexander Philipov. "Style and Skill: Hedge Funds, Mutual Funds, and Momentum". In: *Management Science* 66.12 (Dec. 2020), pp. 5505–5531.

Event Studies

52 Robert W. Holthausen and Richard W. Leftwich. "The effect of bond rating changes on common stock prices". In: *Journal of Financial Economics* 17.1 (Sept. 1986), pp. 57–89.

53 Victor L. Bernard and Jacob K. Thomas. "Post-Earnings-Announcement Drift: Delayed Price Response or Risk Premium?" In: *Journal of Accounting Research* 27 (1989), pp. 1–36.

54 Victor L. Bernard and Jacob K. Thomas. "Evidence that stock prices do not fully reflect the implications of current earnings for future earnings". In: *Journal of Accounting and Economics* 13.4 (Dec. 1990), pp. 305–340.

55 Michael Salinger. "Standard Errors in Event Studies". In: *Journal of Financial and Quantitative Analysis* 27.1 (Mar. 1992), pp. 39–53.

56 Louis K. C. Chan, Narasimhan Jegadeesh, and Josef Lakonishok. "Momentum Strategies". In: *Journal of Finance* 51.5 (Dec. 1996), pp. 1681–1713.

57 A. Craig MacKinlay. "Event Studies in Economics and Finance". In: *Journal of Economic Literature* 35.1 (Mar. 1997), pp. 13–39.

Behavioral Finance

58 Kent Daniel and Sheridan Titman. "Evidence on the Characteristics of Cross Sectional Variation in Stock Returns". In: *Journal of Finance* 52.1 (Mar. 1997), pp. 1–33.

59 Kent Daniel, David A. Hirshleifer, and Avanidhar Subrahmanyam. "Investor Psychology and Security Market Under- and Overreactions". In: *Journal of Finance* 53.6 (Dec. 1998), pp. 1839–1885.

60 Kent Daniel, David A. Hirshleifer, and Avanidhar Subrahmanyam. "Overconfidence, arbitrage and equilibrium asset pricing". In: *Journal of Finance* 56.3 (June 2001), pp. 921–965.

61 David Hirshleifer and Siew Hong Teoh. "Limited attention, information disclosure, and financial reporting". In: *Journal of Accounting and Economics* 36.1-3 (2003), pp. 337–386.

62 Malcolm Baker and Jeffrey Wurgler. "Investor Sentiment and the Cross-Section of Stock Returns". In: *Journal of Finance* 61.4 (Aug. 2006), pp. 1645–1680.

63 David Hirshleifer, Sonya Seongyeon Lim, and Siew Hong Teoh. "Driven to Distraction: Extraneous Events and Underreaction to Earnings News". In: *The Journal of Finance* 64.5 (Sept. 2009), pp. 2289–2325.

64 David Hirshleifer, Sonya S. Lim, and Siew Hong Teoh. "Limited Investor Attention and Stock Market Misreactions to Accounting Information". In: *The Review of Asset Pricing Studies* 1.1 (Dec. 2011), pp. 35–73.

65 Robert F. Stambaugh, Jianfeng Yu, and Yu Yuan. "The Short of It: Investor Sentiment and Anomalies". In: *Journal of Financial Economics* 104.2 (May 2012), pp. 288–302.

66 Robert F. Stambaugh, Jianfeng Yu, and Yu Yuan. "The long of it: Odds that investor sentiment spuriously predicts anomaly returns". In: *Journal of Financial Economics* 114.3 (Dec. 2014), pp. 613–619.